1. A class consists of 40 students who have secured different grades in an examination. The grades are

classified as F, E, D, C, B, A, and S, where F represents the lowest grade and S represents the highest grade.

The number of students who secured each grade is given below.

Grade Number of Students

F 3

E 4

D 5

C 8

B 9

A 7

S 4

Total 40

Develop a Python program to perform the following tasks:

1. Store the grades and corresponding number of students in suitable data structures.

2. Draw a pie chart to represent the percentage distribution of students across the seven grades.

3. Display the grade labels and percentage values on the pie chart.

4. Draw a bar chart to compare the number of students who secured each grade.

5. Label the X-axis as Grades and the Y-axis as Number of Students.

6. Display the exact number of students above each bar.

7. Add suitable chart titles, legends, and axis labels.

8. Identify the grade secured by the maximum and minimum number of students.

9. Display the percentage of students who secured grades A and S combined.

Code

```
import matplotlib.pyplot as plt


grades = ['F', 'E', 'D', 'C', 'B', 'A', 'S']

students = [3, 4, 5, 8, 9, 7, 4]

total = sum(students)
```

```
# Pie chart
plt.figure(figsize=(7, 5))
plt.pie(students, labels=grades, autopct='%1.1f%%',
        startangle=90)
plt.title("Percentage Distribution of Grades")
plt.legend(title="Grades")
plt.show()

# Bar chart
plt.figure(figsize=(7, 5))
bars = plt.bar(grades, students, color='skyblue',
               label='Students')

plt.xlabel("Grades")
plt.ylabel("Number of Students")
plt.title("Number of Students in Each Grade")
plt.legend()

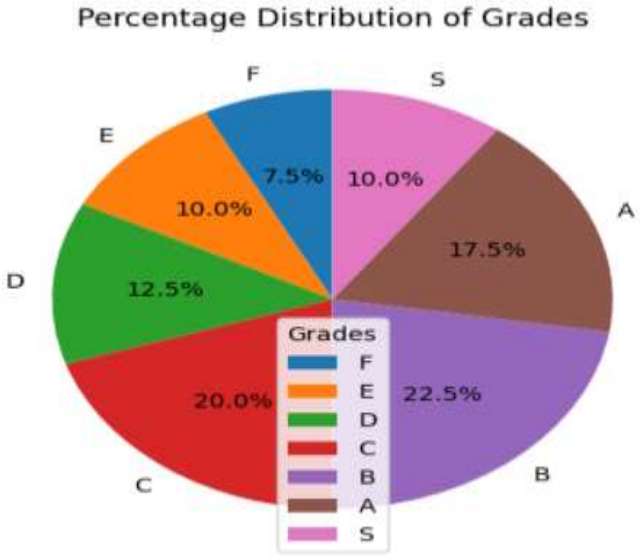
for bar in bars:
    y = bar.get_height()
    plt.text(bar.get_x() + bar.get_width() / 2,
             y + 0.1, int(y), ha='center')

plt.show()

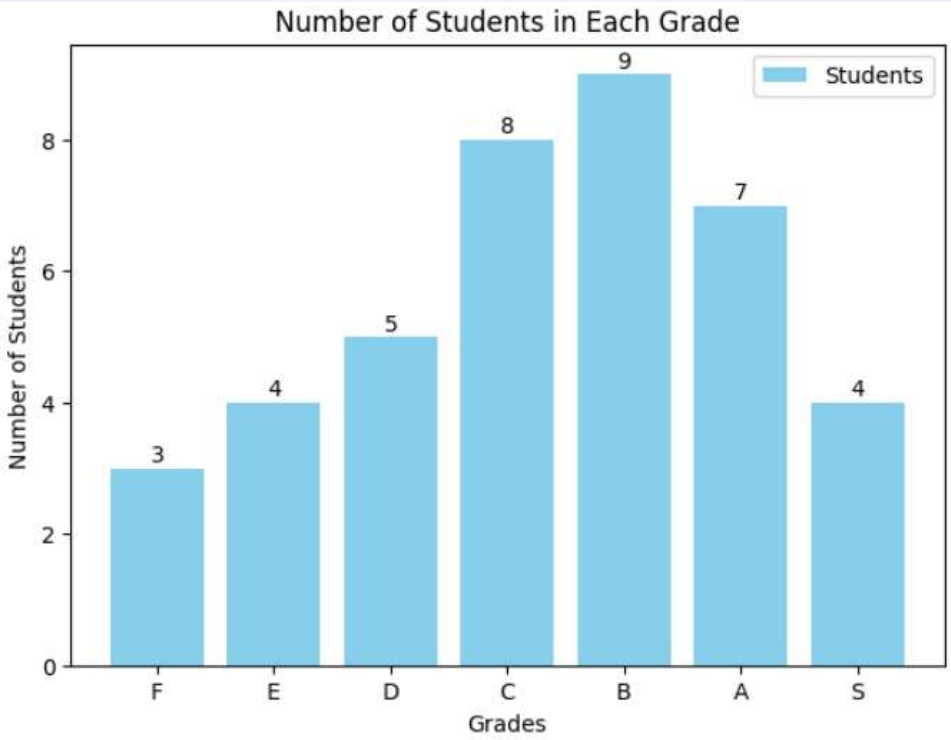
# Maximum and minimum
print("Maximum:", grades[students.index(max(students))],
      "-", max(students), "students")
print("Minimum:", grades[students.index(min(students))],
      "-", min(students), "students")
```

```
# Combined percentage of A and S
combined = students[5] + students[6]
print("A and S combined percentage:",
    combined / total * 100, "%")
```

output



Number of Students in Each Grade




```
Maximum: B - 9 students
Minimum: F - 3 students
A and S combined percentage: 27.500000000000004 %
```